

International Institute of Information Technology, Hyderabad

SC1.440 Dynamical Processes in Complex Networks

Opinion Network Formation

DPCN Assignment 1

Team 34

Kushal Balabhadruni

Rohit Jeswanth

Vishak Kashyap K

GitHub:

<https://github.com/vishakkashyapk30/dpcn-a1-opinion-network-team34>

Table of Contents

1	Team Name	1
2	GitHub Link	1
3	Dataset Documentation	1
3.1	Survey structure	1
3.2	Encoding and filters	1
3.3	Univariate picture	1
4	Pipeline Followed	1
4.1	Conceptual goal	1
4.2	Agent similarity	2
4.3	Threshold selection	2
4.4	Attitude projection	2
5	Analysis and Visualizations	2
5.1	Agent communities (Louvain)	2
5.2	Girvan–Newman cut	3
5.3	Attitude co-endorsement network	3
5.4	Which statements sit at the core, and which ones bridge	5
5.5	Contested vs. community-organizing items	5
6	Results and Discussion	5
6.1	What the network view adds	6
7	Individual Contribution	7
8	References	7

1 Team Name

Team 34

2 GitHub Link

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Reproduction: `code/01_data_and_network.ipynb`, `code/02_analysis.ipynb`, and `code/regenerate_figures.ipynb`.
Figures used below come from that regeneration notebook and from `figures/cytoscape/`.

3 Dataset Documentation

3.1 Survey structure

The survey records opinions from **96** respondents on **60** five-point Likert items, in four blocks of 15: Technology (T), Education (E), Ethics/Society (S), and Environment (V). Following [MacCarron et al. \(2020\)](#), we treat the cleaned table as a **bipartite** participant–item incidence structure. Person–person and statement–statement graphs are projections of that bipartite view, not contact graphs.

3.2 Encoding and filters

Response labels map to $[-1, -0.5, 0, 0.5, 1]$. Blanks and **No Comments** are missing (not Neutral). Respondents with under **50%** item completeness are dropped, leaving **87** of 96. Pairwise agreement scores require at least 20 jointly answered items.

3.3 Univariate picture

Figure 1 shows a strong Agree skew: mass concentrates at 0.5 and 1.0, with few strong negatives. Domain violins confirm the same lean in every theme, but not equally. Environment is the most cohesive positive block (mean 0.69, variance 0.14); Society follows (mean 0.64). Technology is middling; Education is softest and noisiest (mean 0.45, variance 0.33). Averages alone say “the class mostly agrees.” The network pipeline asks a sharper question: among agreeable respondents, who still clusters with whom, and which items create those clusters?

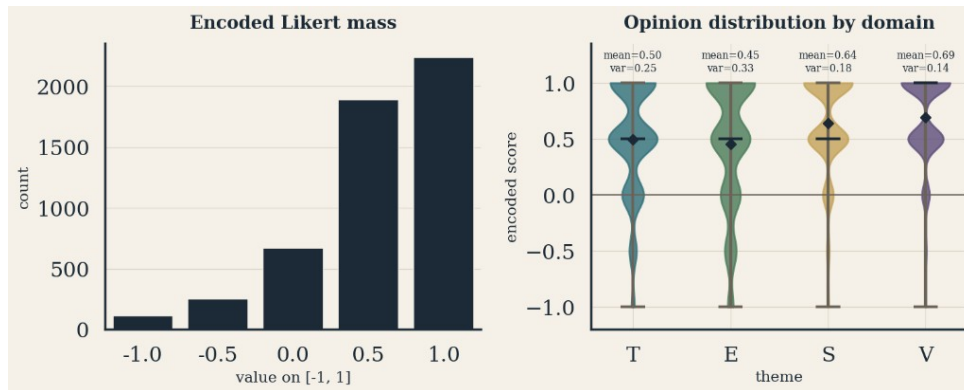


Figure 1: Encoded Likert mass (left) and opinion distributions by theme (right).

4 Pipeline Followed

4.1 Conceptual goal

Following [MacCarron et al. \(2020\)](#), [Dinkelberg et al. \(2021\)](#), and [Maher et al. \(2020\)](#), we build two complementary graphs: an **agent agreement network** (nodes = respondents; edges = high answer similarity) and an **attitude co-endorsement network** (nodes = statements; edges = joint positive endorsement). These are homophily / belief networks.

4.2 Agent similarity

For jointly answered items F_{ij} with $n_{ij} = |F_{ij}|$,

$$S_{ij} = n_{ij} - \sum_{f \in F_{ij}} |q_{if} - q_{jf}|.$$

Larger S_{ij} means closer answer profiles. The empirical S_{ij} distribution (Figure 2, left) peaks near 35 and is left-skewed; most pairs are only moderately similar.

4.3 Threshold selection

We retain edges with $S_{ij} \geq \theta$ and choose the largest θ such that the giant component still covers about **80%** of respondents (Dinkelberg et al., 2021). Figure 2 (right) shows that the giant fraction stays near 1 until roughly $\theta \approx 20$, then declines; it crosses the 0.8 target at $\theta = 41.5$. That cut sits in the upper tail of S_{ij} , so the kept edges are the strongest agreements only. The resulting agent graph has 87 nodes, 456 edges, density 0.122, average degree ≈ 10.5 , and clustering 0.45.



Figure 2: Left: pairwise agreement S_{ij} with chosen $\theta = 41.5$. Right: giant-component fraction vs. threshold (target 80%).

4.4 Attitude projection

Statement pairs use co-endorsement $W_{ab} = \sum_i q_{ia}q_{ib}$ on the $[-1, 1]$ encoding. Strong positive W_{ab} means the two items are routinely affirmed together. We keep high- W edges for visualization and report the top twenty pairs in Table 1.

5 Analysis and Visualizations

Network drawings are Cytoscape.js force-directed views (cose layout; edge weight controls spring strength). Interactive HTML lives under `figures/cytoscape/`.

5.1 Agent communities (Louvain)

Louvain returns 20 communities in total, but only three contain more than one member: sizes **37** (C0), **29** (C1), and **4** (C2). Modularity is moderate ($Q \approx 0.275$), meaning these communities are only somewhat more internally connected than a random graph of the same density would be. The other 17 communities are singletons, and they are exactly the respondents who have zero edges at $\theta = 41.5$ (cleared the completeness filter but did not agree strongly enough with anyone else): nodes with no edges cannot be merged into any group, so Louvain correctly leaves them alone. In Figure 3, C0 and C1 sit next to each other with direct edges running between their members (they are not two separate components); C2 sits at the boundary between them; the singleton respondents appear as isolated points scattered around the frame.

Figure 4 plots each community’s mean score on each theme. Both C0 and C1 score above zero (the Agree side) on all four themes, so the two communities are not an agree-versus-disagree split. C0 scores higher than C1 on every theme, most noticeably on Society (mean ≈ 0.82) and Environment (mean ≈ 0.87); C1’s four theme means stay in

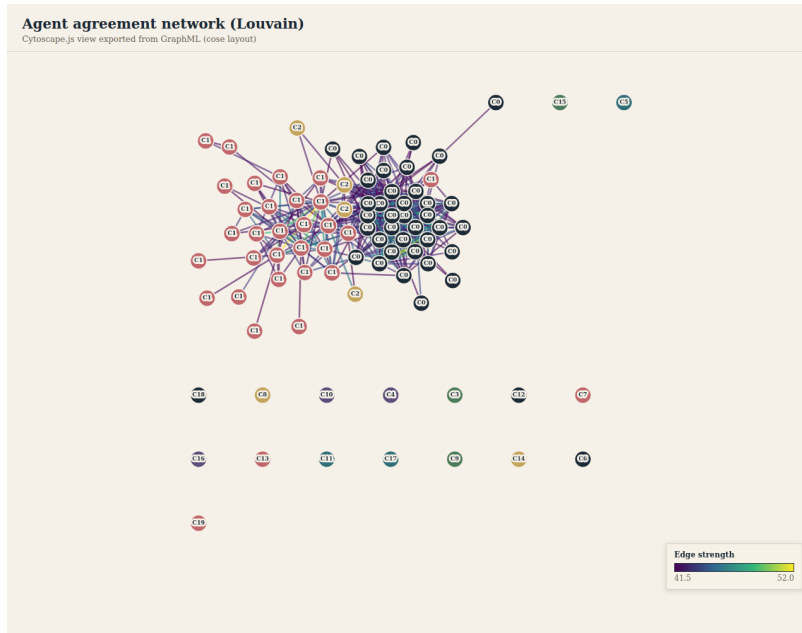


Figure 3: Agent agreement network colored by Louvain community (Cytoscape.js, force-directed by edge weight).

a narrower 0.42–0.56 range. C2 ($n = 4$) has moderate Technology, Education, and Society means alongside a higher Environment mean. Education has the lowest mean in all three communities. So C0 and C1 differ in how strongly and consistently their members agree, not in which direction they lean.

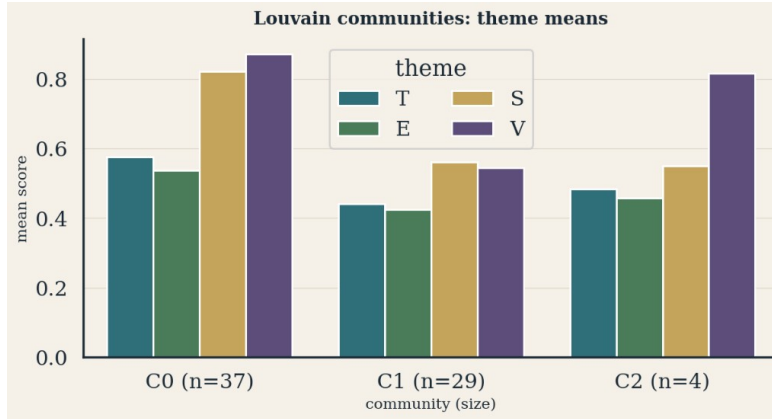


Figure 4: Mean theme scores for Louvain communities with size at least 3.

5.2 Girvan–Newman cut

Girvan–Newman repeatedly deletes the edge with the highest *betweenness* (the edge lying on the largest number of shortest paths between other node pairs) until the graph splits. Applied to the 70-node giant component, the first split is highly uneven: **64, 4, and 2** nodes (Figure 5), not two groups of comparable size. The highest-weight edges (yellow) sit inside the 64-node group, and the two small splinter groups are attached to it by only one or two low-weight edges, which is exactly why the algorithm removed those edges first. Together with Louvain’s moderate modularity, this points to one dominant, densely connected group with a few small groups loosely attached, rather than two comparably sized opposing camps as in the bipolar ANES-style splits reported in the cited literature.

5.3 Attitude co-endorsement network

Figure 6 shows the statement pairs whose co-endorsement W_{ab} clears the 70th-percentile threshold. Society and Environment items form a densely connected core with many edges among them; several Education and Technology items (e.g. E02–E04, E09, E12, T02, T08, T09) have no surviving edge and appear as isolated nodes. Edges do

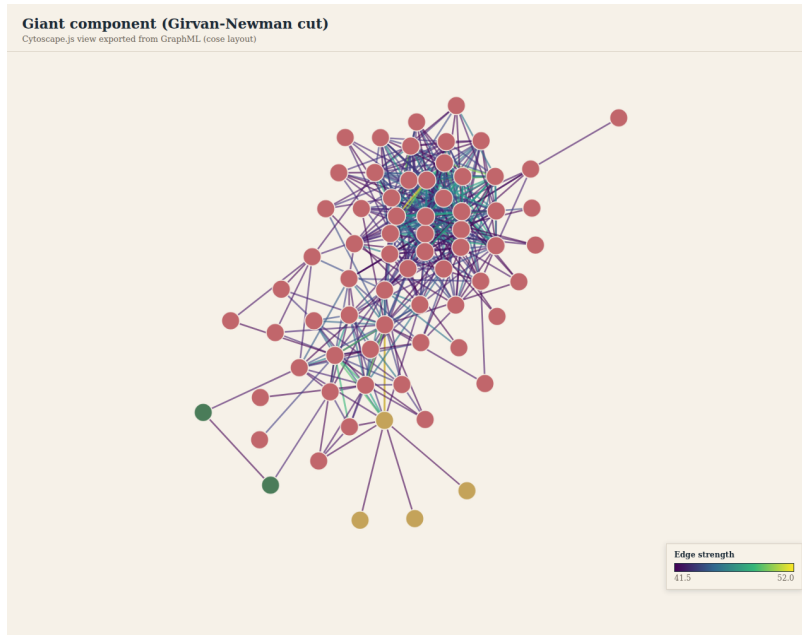


Figure 5: Giant component colored by the Girvan–Newman cut (Cytoscape.js).

connect different themes, but the core is dominated by Society and Environment items together with two broadly endorsed Education items, continuous learning (E15) and curriculum update (E11), rather than by an even mix of all four themes.

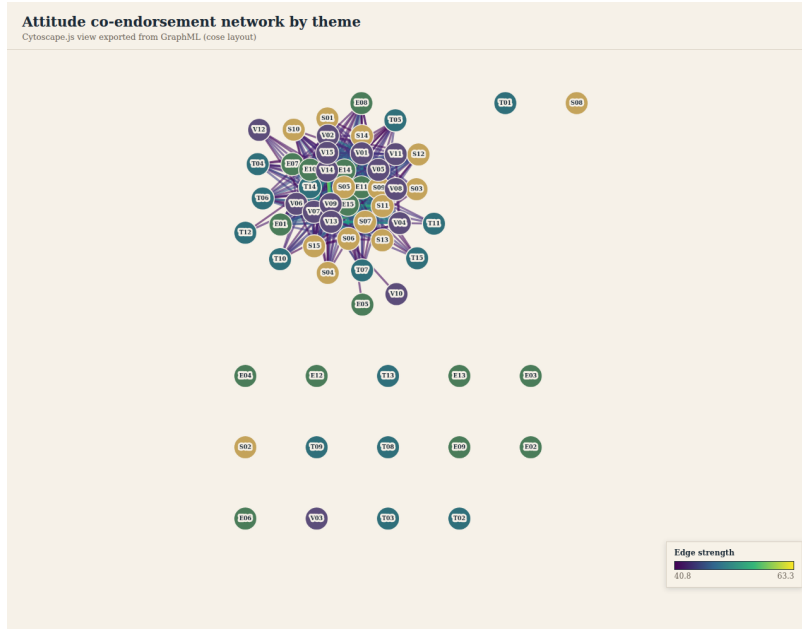


Figure 6: Attitude co-endorsement network colored by theme (Cytoscape.js).

Table 1 lists the twenty strongest co-endorsement pairs. **E15** appears in 9 of these 20 pairs, more than any other item, pairing with items from all three other themes; **E11**, **V13**, **V09**, and **S05** also recur repeatedly. Six of the twenty pairs connect two items from the same theme (e.g. E11–E15, E14–E15, V09–V13, V13–V14); the remaining fourteen connect two different themes (e.g. E15–V13, E15–S05, S05–V13). In other words, respondents who strongly endorse one broadly worded statement, such as E15 (“continuous learning is essential”), tend to also strongly endorse the other broadly endorsed statements regardless of theme label. None of the contested Education items identified in the next subsection (E02–E04) appears anywhere in this top-20 list.

Table 1: Top 20 attitude co-endorsement edges (W_{ab}).

#	A	B	Short A	Short B	W
1	E11	E15	Curricula updated for emerging tech	Continuous learning essential	63.25
2	E15	V13	Continuous learning essential	Firms accountable for env. impact	62.00
3	E15	S05	Continuous learning essential	Control over personal data	61.75
4	E14	E15	University-industry collaboration	Continuous learning essential	61.50
5	V09	V13	Design for reuse/repair	Firms accountable for env. impact	61.25
6	E15	V09	Continuous learning essential	Design for reuse/repair	61.00
7	S05	V13	Control over personal data	Firms accountable for env. impact	60.50
8	E10	E15	Innovation over rote learning	Continuous learning essential	60.50
9	E15	S11	Continuous learning essential	Free expression (non-harmful)	60.00
10	E15	V14	Continuous learning essential	Intl. env. cooperation	59.50
11	E11	S05	Curricula updated for emerging tech	Control over personal data	59.50
12	E11	V09	Curricula updated for emerging tech	Design for reuse/repair	59.50
13	V13	V14	Firms accountable for env. impact	Intl. env. cooperation	59.00
14	E11	V13	Curricula updated for emerging tech	Firms accountable for env. impact	59.00
15	S05	V09	Control over personal data	Design for reuse/repair	58.50
16	E10	V13	Innovation over rote learning	Firms accountable for env. impact	58.25
17	V09	V14	Design for reuse/repair	Intl. env. cooperation	58.00
18	E11	S11	Curricula updated for emerging tech	Free expression (non-harmful)	57.75
19	E15	V15	Continuous learning essential	Protect resources for future gens	57.50
20	E10	V09	Innovation over rote learning	Design for reuse/repair	57.25

5.4 Which statements sit at the core, and which ones bridge

Degree alone only counts how many edges a node has; two other centrality measures ask more specific questions about the same attitude network. **Eigenvector centrality** scores a statement highly if it connects to other well-connected statements, computed by repeatedly averaging each node’s score with its neighbors’ scores until the ranking stops changing. **Betweenness centrality** scores a statement highly if it lies on many shortest paths between other pairs of statements, i.e. removing it would make many other items harder to reach from each other through the network.

The top eigenvector-centrality items are **E15** (continuous learning, 0.230), **V13** (corporate environmental accountability, 0.225), **V09** (reuse/repair design, 0.219), **E11** (curriculum update, 0.219), and **S05** (personal-data control, 0.219); these are the same five items that dominate Table 1, confirming that they sit at the well-connected heart of the network rather than merely appearing in a few strong pairs. The lowest eigenvector scores belong almost entirely to Technology and Education items (T13, E02, E03, E04, T03, T02, T08, T01), the same items that are disconnected in Figure 6.

Betweenness centrality tells a different story: **T14** (social media algorithms cause polarization) has the highest betweenness (0.0452), nearly 50% higher than the runner-up V13 (0.0301); E15 (0.0245), V07 (0.0245), and S11 (0.0212) follow. T14’s eigenvector score (0.190) is noticeably lower than the other top-betweenness items, so T14 is not itself part of the densely-interconnected core; instead it is the single most important bridge connecting the more loosely-attached Technology items to that core. V13, E15, V07, and S11 score highly on both measures, meaning they are central in two different senses at once: deeply embedded *and* load-bearing bridges.

5.5 Contested vs. community-organizing items

Variance marks disagreement *content*; shuffle importance ($1 - \overline{\text{ARI}}$ after permuting an item vs. baseline Louvain) marks disagreement that *organizes people* (Figure 7).

The most contested items are Education-led: **E04**, **E03**, **E02**, then Technology item T08 and Society item S02. No Environment item enters the top variance list, consistent with V’s tight violin in Figure 1. Shuffle importance also elevates Education (**E04**, **E01**) and mixes in T02, S15, T14, and several V items. So Environment can still structure communities through *intensity* differences (C0 vs. C1) even when raw variance is low, while Education items are both noisy and structurally decisive.

6 Results and Discussion

Putting the pieces together yields a coherent class portrait.

1. **One connected majority, not separate camps.** Likert mass and theme violins show broad Agree-side scores

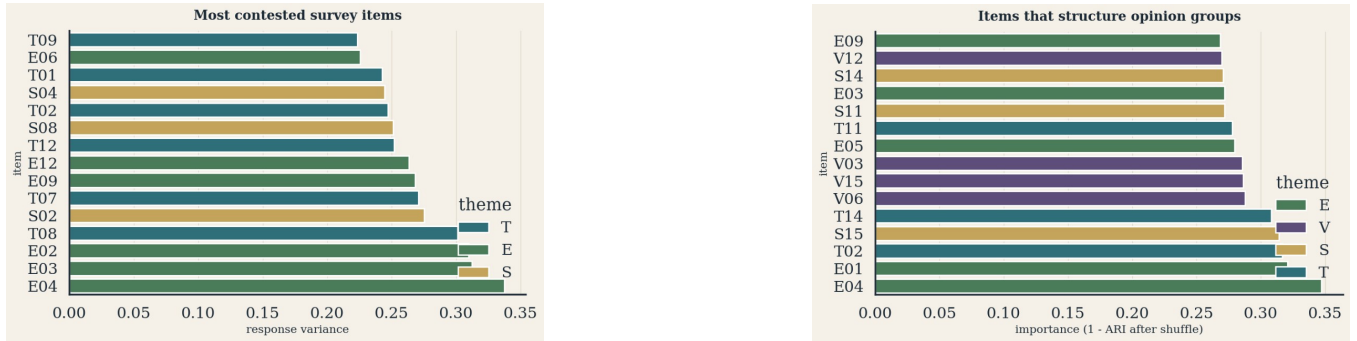


Figure 7: Left: highest-variance (most contested) items. Right: items whose shuffle most disrupts Louvain communities.

across the class. At $\theta = 41.5$, 70 of 87 respondents ($\sim 80\%$) form a single giant component, connected through high-agreement edges; the strongest co-endorsement edges among statements (Table 1, Figure 6) connect mostly Society and Environment items.

- Two neighborhoods that differ in strength, not direction.** Louvain’s two large communities (C0, 37 members; C1, 29 members) are both positive on every theme, but C0’s means run higher, especially on Society and Environment. Girvan–Newman’s uneven 64/4/2 split, together with Louvain’s moderate modularity ($Q = 0.275$), rules out a two-camp, opposed-ideology structure.
- Education items account for most of the disagreement.** Education has the lowest mean and highest variance among the four themes; its items dominate the contested-item ranking and the structural-importance ranking (E04, E03, E02; E04, E01) in Figure 7. These same contested items do not appear among the strongest co-endorsement pairs in Table 1, which instead feature the two near-consensus Education items E15 and E11.
- A small set of statements is endorsed together far more than the rest.** The attitude network (Figure 6) and Table 1 show that sustainability items (V09, V13, V14, V15), the data-control item (S05), and the two consensus Education items (E15, E11) are co-endorsed much more strongly than other item combinations. Technology items and the contested Education items sit on the periphery of the network or are disconnected entirely after thresholding.

6.1 What the network view adds

Computing theme means alone would show only that every theme’s average score is positive. It would not show that 80% of respondents form one connected group at a principled agreement threshold; that this group splits further into two neighborhoods differing in agreement strength rather than direction; that some items are contested (high variance) while a partly different set of items is structurally decisive (high shuffle-importance); or that specific statements are co-endorsed together far more than others. The MacCarron/Dinkelberg pipeline recovers all of this by turning the Likert table into a graph that can be thresholded, partitioned, and stress-tested: students share a Society/Environment agreement core, while education practice (and related Technology items) account for most of the remaining disagreement.

A practical reading for the class is therefore not “students disagree about the environment,” but “students largely co-endorse sustainability and data-rights statements, and diverge more when asked how universities and AI should teach.” That distinction is visible only once agreement edges, community profiles, contestation, and co-endorsement hubs are read together.

7 Individual Contribution

Member	Contribution
Kushal Balabhadruni	Section 3: designed the $[-1, 0.5, 0, 0.5, 1]$ Likert encoding and the missing-vs.-neutral rule for blanks/ No Comments ; applied the 50% completeness filter ($96 \rightarrow 87$ respondents); computed the per-theme means/variances and built Figure 1 (encoded Likert mass and theme violins).
Rohit Jeswanth	Section 4 and Section 5.1, 5.2, 5.5: derived the agreement score S_{ij} and ran the threshold scan that fixed $\theta = 41.5$ against the 80% giant-component target (Figure 2); computed Louvain modularity ($Q = 0.275$) and the Girvan–Newman betweenness cut (64/4/2 split); exported the agent-network GraphML used for Figures 3–5; ran the response-variance and shuffle-based item-importance rankings behind Figure 7.
Vishak Kashyap K	Section 4.4 and Section 5.3–5.4: defined the co-endorsement weight W_{ab} and built the attitude network and Table 1 top-20 ranking; computed eigenvector and betweenness centrality on the attitude network; produced the Cytoscape.js force-directed renderings (Figures 3, 5, 6); assembled the LaTeX report and repository layout/README.

8 References

- MacCarron, P., Maher, P. J., and Quayle, M. (2020). Identifying opinion-based groups from survey data: a bipartite network approach. arXiv:2012.11392.
- Dinkelberg, A., MacCarron, P., Maher, P. J., and Quayle, M. (2021). Detect opinion-based groups and reveal polarisation in survey data. arXiv:2104.14427.
- Maher, P. J., MacCarron, P., and Quayle, M. (2020). Mapping public health responses with attitude networks. *British Journal of Social Psychology*.